

OPERATORS: Operator is a special symbol designed for a particular task [work]. C comes with 44 operators and 14 separators. Operator works on operands. Based on **no of operands** participating in operation, the operators divided into **3 types**.

1. **Unary operator:** Require one operand.
Eg: `a++`, `a--`, `++a`, `--a`, `sizeof(a)`, `~a`, `!a`, `+a`, `-a`,...
2. **Binary operator:** Require two operands.
Eg: `a+b`, `a>b`, `a<=b`, `a!=b`, `a<<b`,....
3. **Ternary / conditional operator [?:]:** Require 3 operands.
Eg: **conditional part ? true part : false part ;**

`5 > 9 ? " 5 is big " : " 9 is big " ;`

`7 > 2 ? "7 is big" : " 2 is big " ;`

Based on operation, the operators divided into several types.

1. **Assignment operator [=]:** It copies the value on its right side into the variable on its left side. In assignment operation, the **left side operand should be a variable**. i.e. expressions and constants not allowed on left side.

Eg:

`a=100;`

`b=1.2;`

`c='x';`

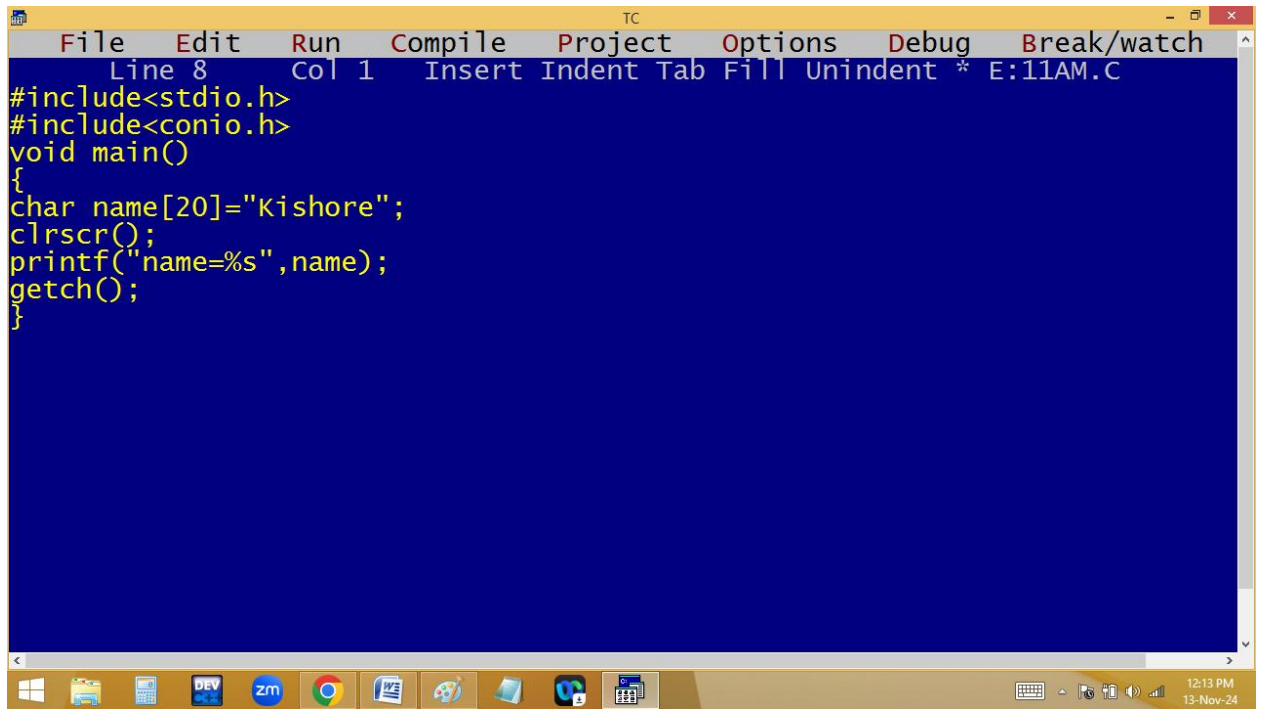
`d="abc";` ➔ Error ➔ string copy not allowed with =

The image shows a screenshot of the Turbo C++ (TC) IDE. The top window is the source code editor, which has a blue background and yellow text. It contains a C program that includes `<stdio.h>` and `<conio.h>`. The `main` function declares variables `a` (int), `b` (float), and `c` (char). It initializes `a` to 100, `b` to 1.2, and `c` to 'X'. It then uses `printf` to print the values of `a`, `b`, and `c`, followed by a call to `getch()`. The status bar at the top of the editor shows "Line 10 Col 6" and "Insert Indent Tab Fill Unindent * E:11AM.C".

The bottom window is the output console, which has a black background and white text. It displays the output of the program: `a=100, b=1.200000, c=X_`. The status bar at the bottom of the console shows "TC".

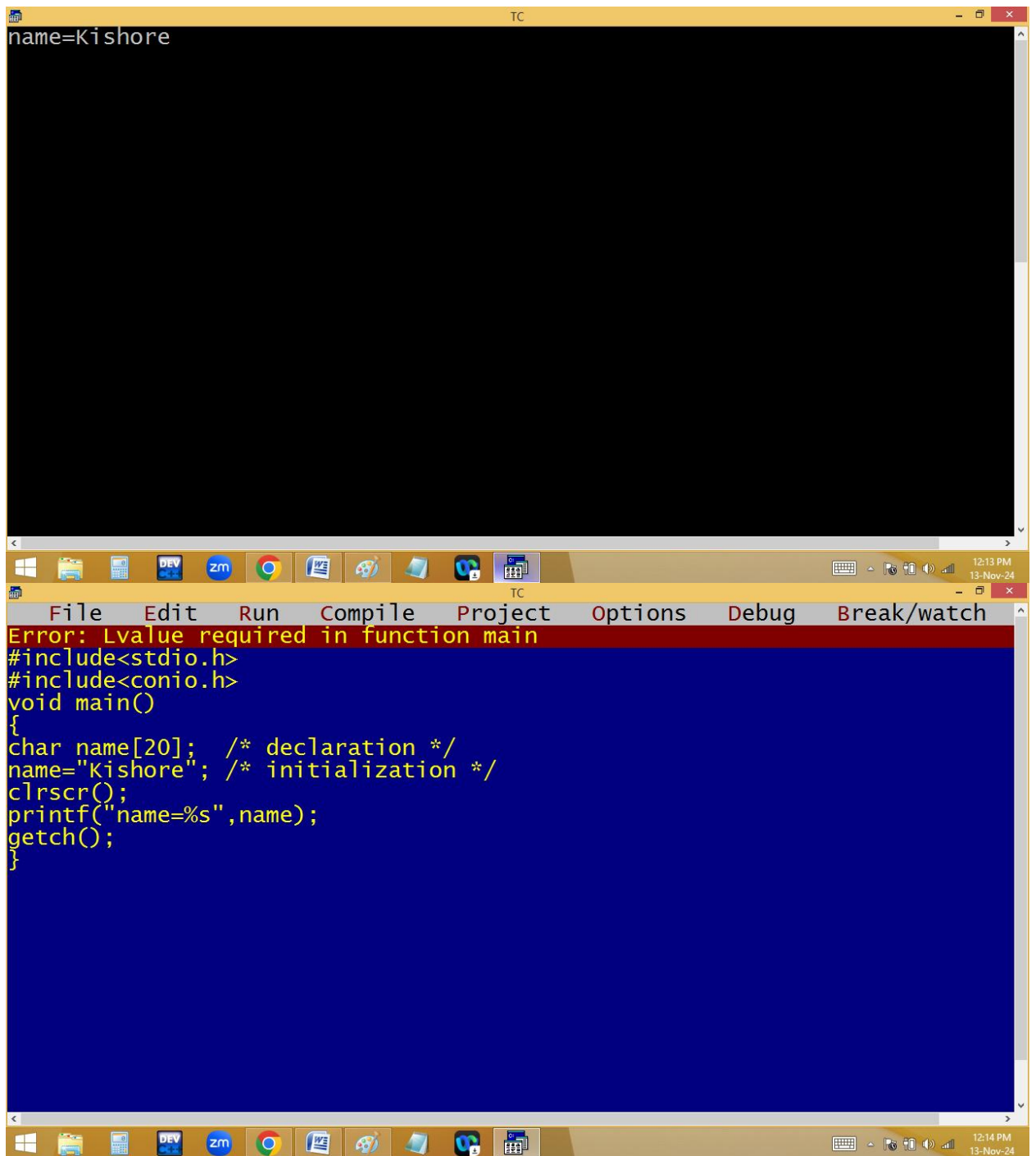
```
File Edit Run Compile Project Options Debug Break/watch
Line 10 Col 6 Insert Indent Tab Fill Unindent * E:11AM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a;
float b;
char c;
clrscr();
a=100;
b=1.2;
c='X';
printf("a=%d, b=%f, c=%c",a,b,c);
getch();
}
```

a=100, b=1.200000, c=X_



```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 8 Col 1 Insert Indent Tab Fill Unindent * E:11AM.C
#include<stdio.h>
#include<conio.h>
void main()
{
char name[20]="Kishore";
clrscr();
printf("name=%s",name);
getch();
}
```

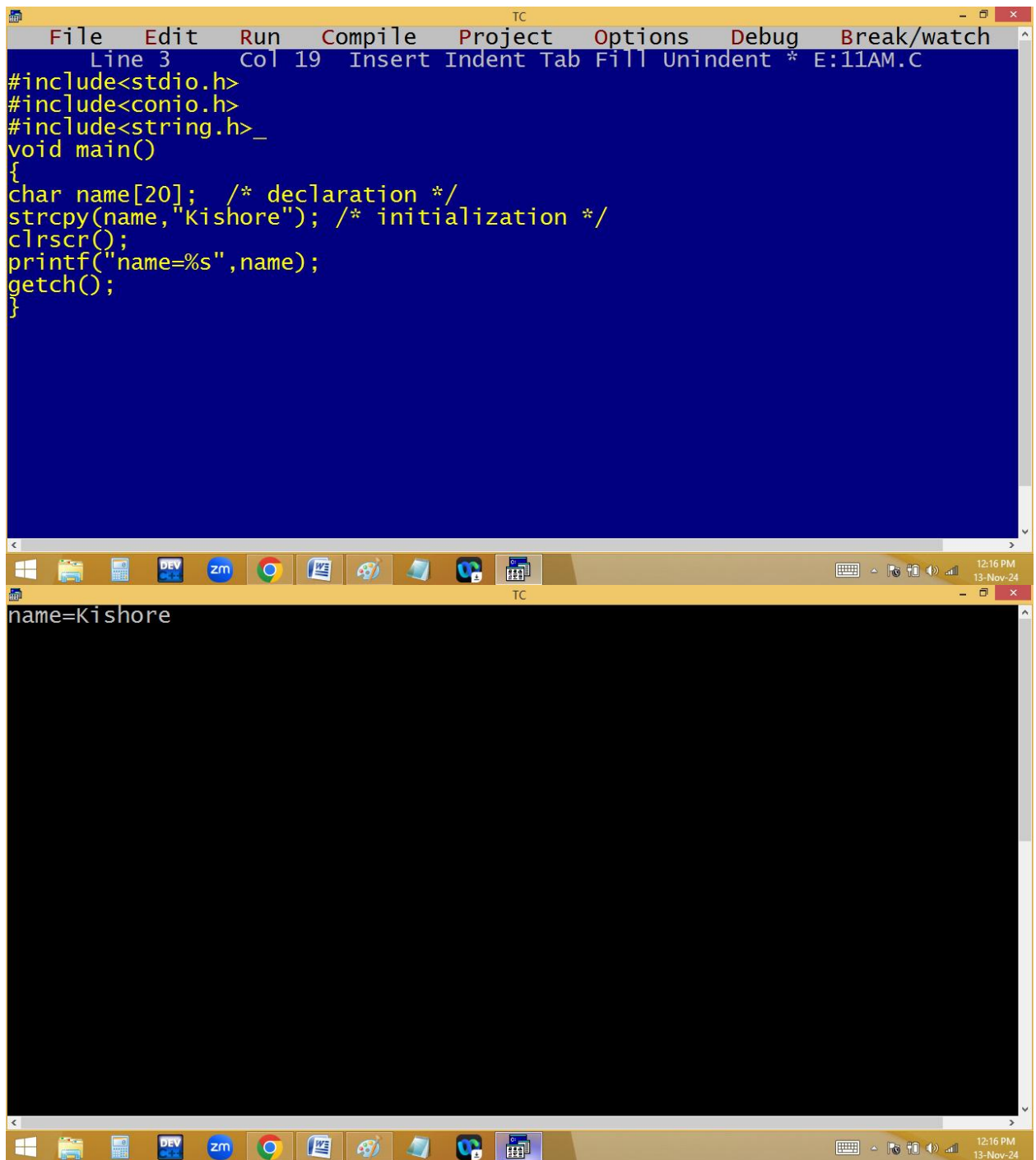
12:13 PM
13-Nov-24



The image shows a screenshot of the Turbo C++ (TC) IDE. The top window, titled 'TC', displays the text 'name=Kishore'. The bottom window, also titled 'TC', shows the source code and a compilation error. The error message, 'Error: Lvalue required in function main', is highlighted in red. The source code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
char name[20]; /* declaration */
name="Kishore"; /* initialization */
clrscr();
printf("name=%s",name);
getch();
}
```

The IDE's menu bar includes File, Edit, Run, Compile, Project, Options, Debug, and Break/watch. The Windows taskbar at the bottom shows the time as 12:13 PM and 12:14 PM on 13-Nov-24.

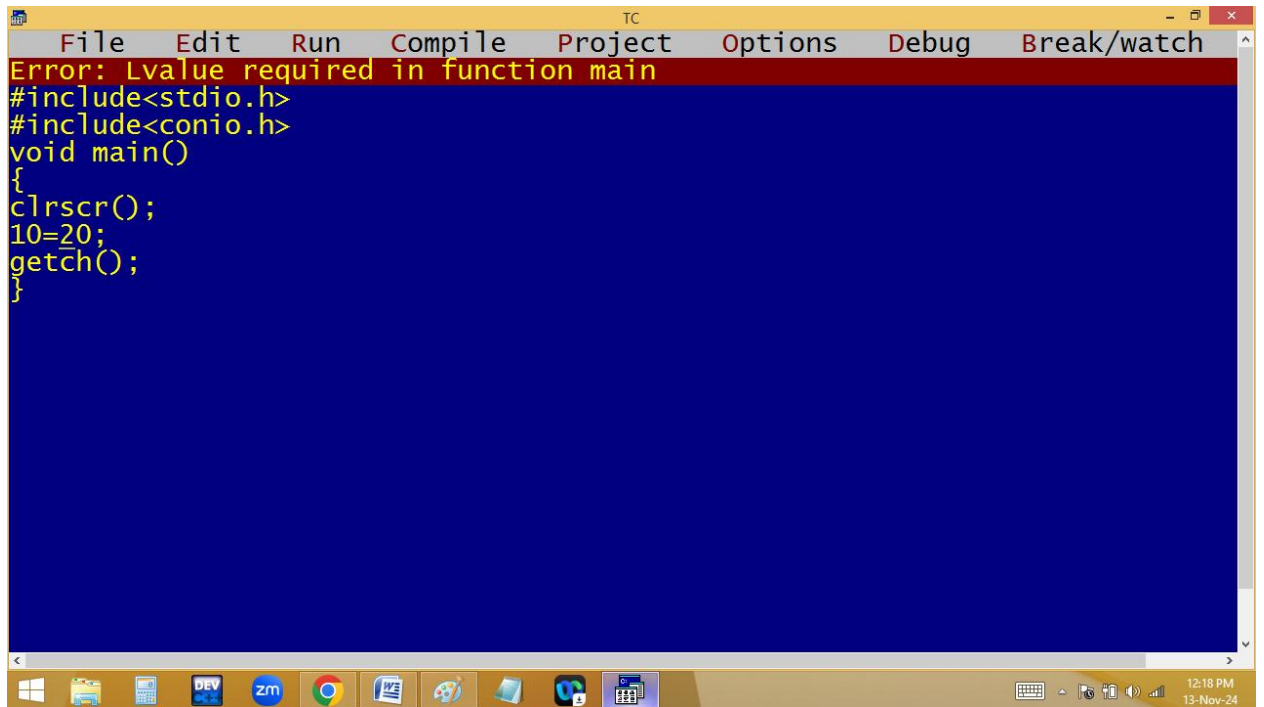


The image shows a screenshot of the Turbo C++ (TC) IDE. The top window displays a C program with the following code:

```
File Edit Run Compile Project Options Debug Break/watch
Line 3 Col 19 Insert Indent Tab Fill Unindent * E:11AM.C
#include<stdio.h>
#include<conio.h>
#include<string.h>_
void main()
{
char name[20]; /* declaration */
strcpy(name,"Kishore"); /* initialization */
clrscr();
printf("name=%s",name);
getch();
}
```

The bottom window shows the output of the program, which is "name=Kishore". The Windows taskbar at the bottom indicates the time is 12:16 PM on 13-Nov-24.

10=20; ➔ Error

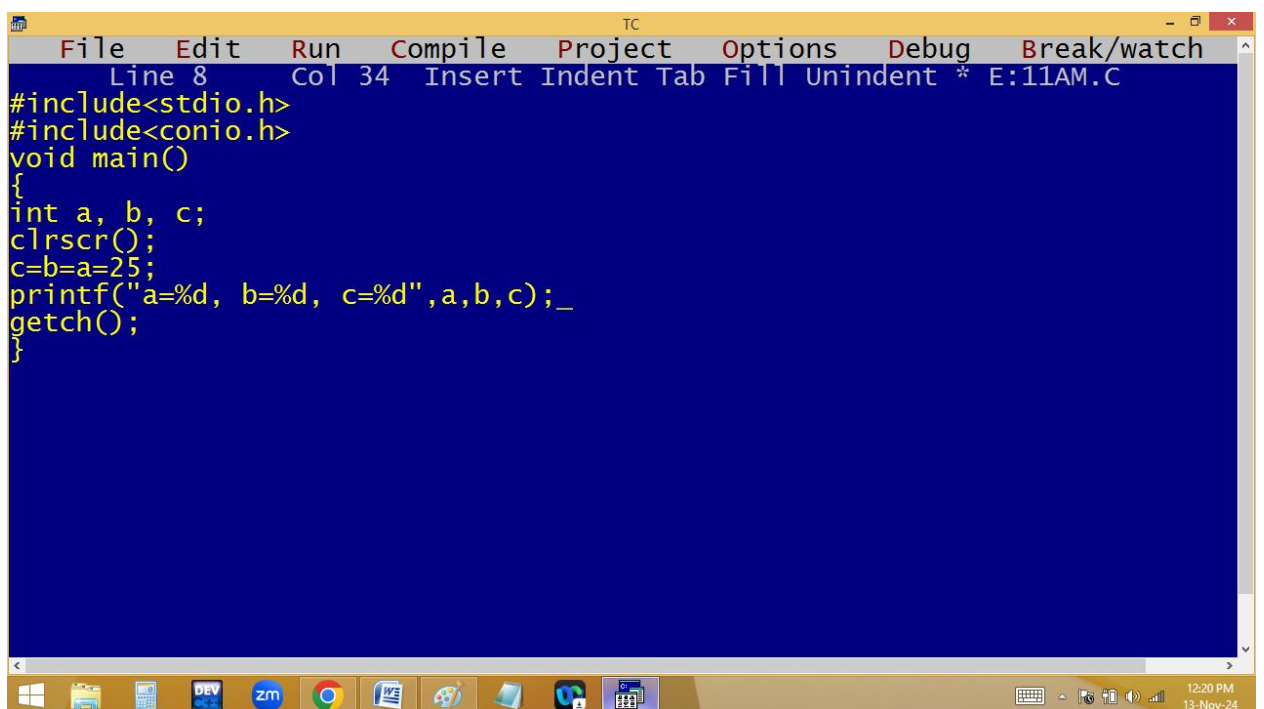


The screenshot shows the Turbo C++ (TC) IDE interface. The menu bar includes File, Edit, Run, Compile, Project, Options, Debug, and Break/watch. A red error message banner at the top reads "Error: Lvalue required in function main". The code editor contains the following code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
10=20;
getch();
}
```

The taskbar at the bottom shows various application icons and the system clock indicating 12:18 PM on 13-Nov-24.

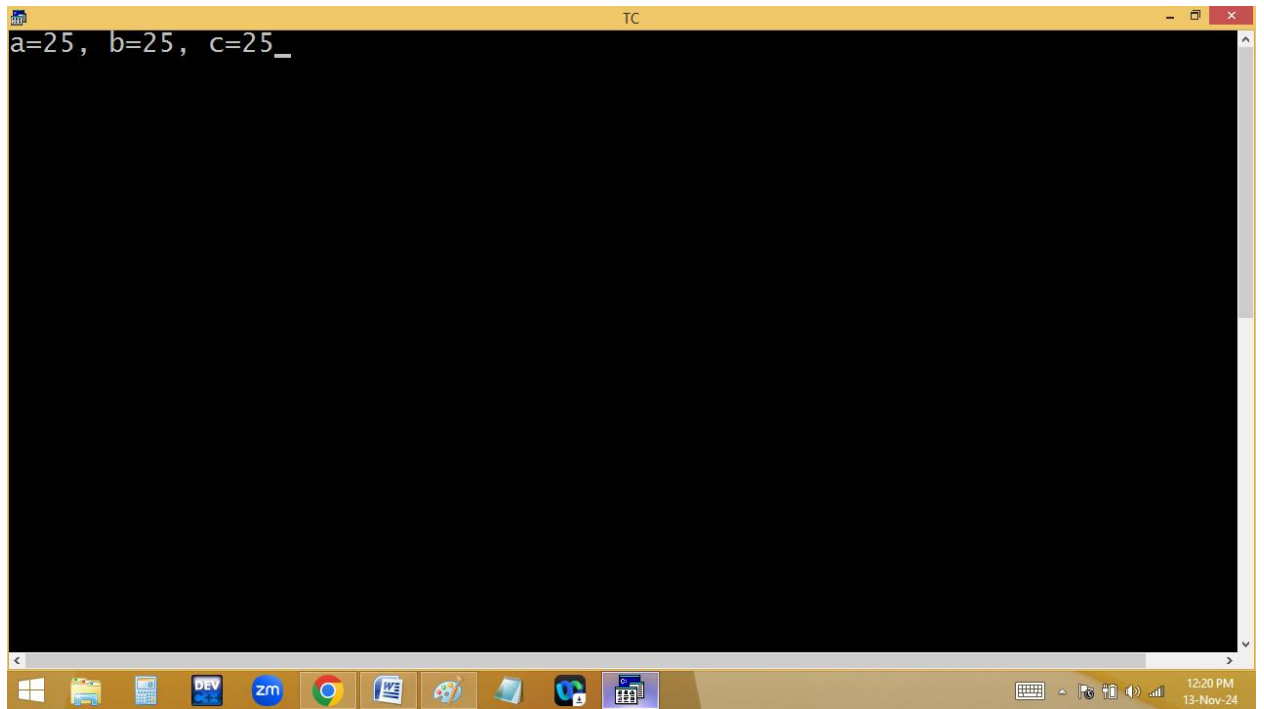
a=b=c=25;



The screenshot shows the Turbo C++ (TC) IDE interface with the same menu bar. The status bar at the top indicates "Line 8 Col 34 Insert Indent Tab Fill Unindent * E:11AM.C". The code editor contains the following code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a, b, c;
clrscr();
c=b=a=25;
printf("a=%d, b=%d, c=%d",a,b,c);_
getch();
}
```

The taskbar at the bottom shows various application icons and the system clock indicating 12:20 PM on 13-Nov-24.



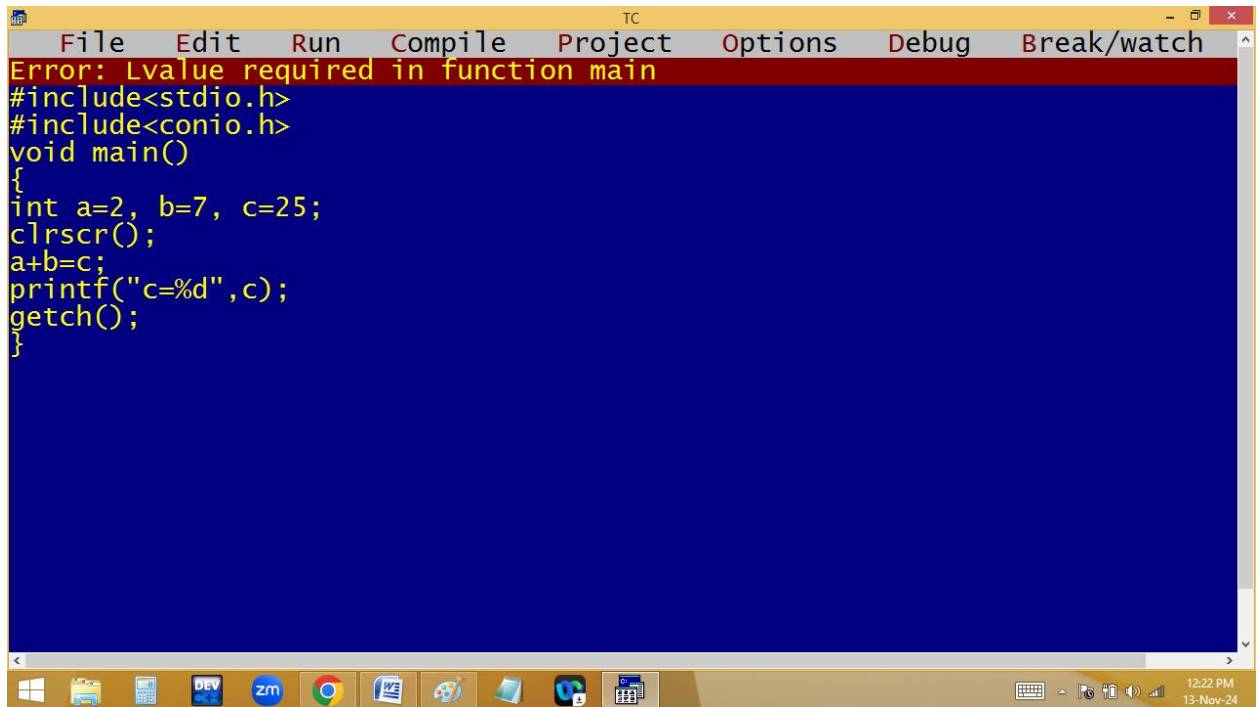
c=2+7 ; → c=9

2+7= 9; → Error

The image shows a screenshot of the Turbo C++ (TC) IDE. The top window is the source code editor, which has a blue background and yellow text. It displays a C program that calculates the sum of two numbers, a=2 and b=7, and stores the result in c. The code includes headers for stdio.h and conio.h, and uses clrscr() to clear the screen before printing the result. The bottom window is the output console, which has a black background and white text, showing the result 'c=9'. The Windows taskbar is visible at the bottom of the screen, showing various application icons and the system clock.

```
File Edit Run Compile Project Options Debug Break/watch
Line 8 Col 15 Insert Indent Tab Fill Unindent * E:11AM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a=2, b=7, c;
clrscr();
c=a+b;
printf("c=%d",c);
getch();
}
```

c=9



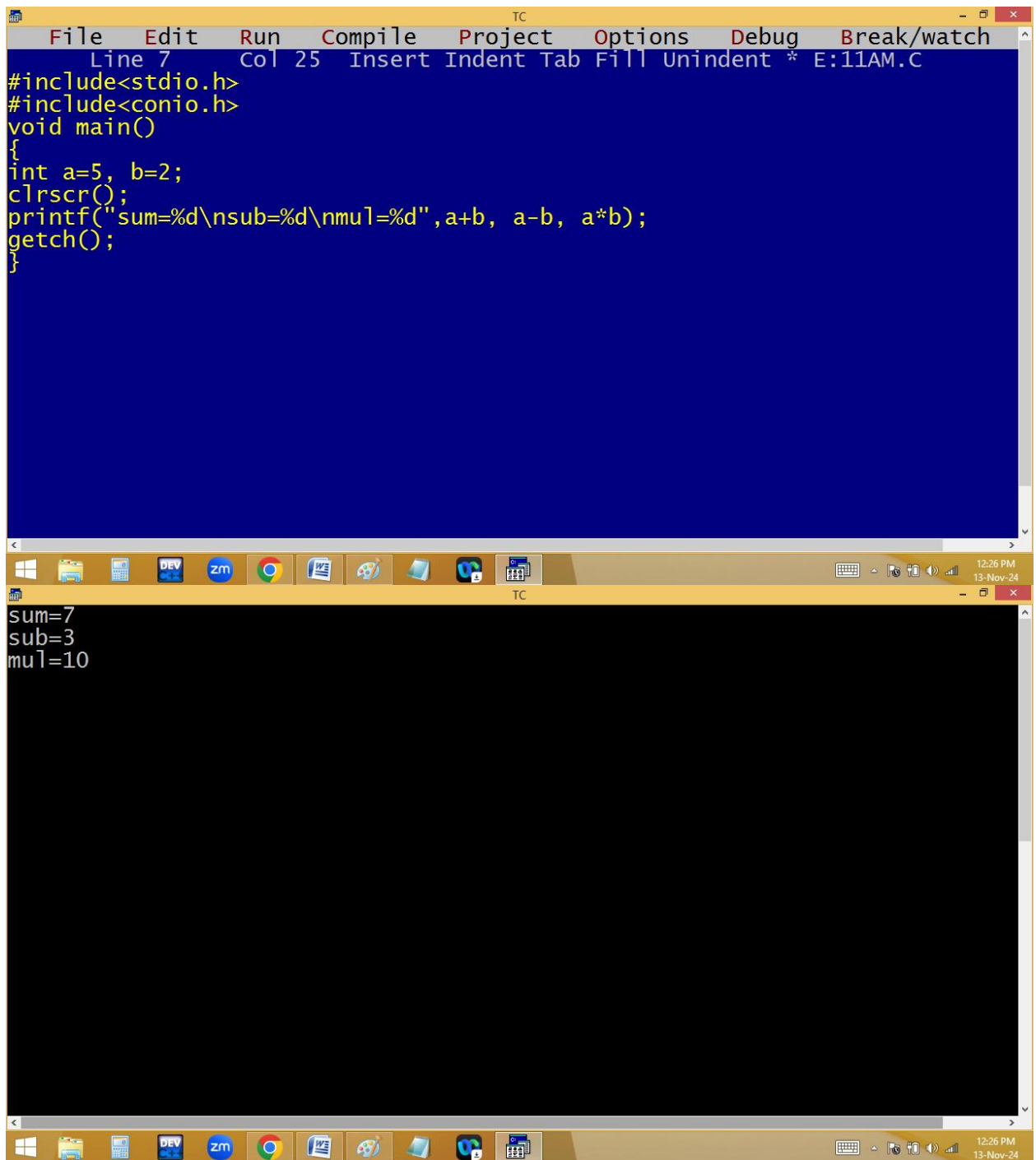
The screenshot shows a Turbo C++ (TC) IDE window. The title bar reads 'TC'. The menu bar includes 'File', 'Edit', 'Run', 'Compile', 'Project', 'Options', 'Debug', and 'Break/watch'. A red error message banner at the top states 'Error: Lvalue required in function main'. The code editor contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a=2, b=7, c=25;
clrscr();
a+b=c;
printf("c=%d",c);
getch();
}
```

The Windows taskbar at the bottom shows various application icons, including Windows Explorer, DEV, zm, Google Chrome, and others. The system clock in the bottom right corner indicates '12:22 PM' and '13-Nov-24'.

2. **Arithmetic operators [+, -, *, %, /]:** They are used for mathematical operations.

Eg: a+b, a-b, a*b,....



The image shows two windows from the Turbo C++ (TC) IDE. The top window is the source code editor, displaying a C program. The code includes `<stdio.h>` and `<conio.h>`, defines `main()`, declares `int a=5, b=2;`, calls `clrscr();`, and uses `printf` to print the sum, difference, and product of `a` and `b`. The bottom window shows the output of the program, which is the result of the `printf` statement: `sum=7`, `sub=3`, and `mul=10`.

```
File Edit Run Compile Project Options Debug Break/watch
Line 7 Col 25 Insert Indent Tab Fill Unindent * E:11AM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a=5, b=2;
clrscr();
printf("sum=%d\nsub=%d\nmul=%d",a+b, a-b, a*b);
getch();
}
```

```
sum=7
sub=3
mul=10
```

% - modules [Remainder]

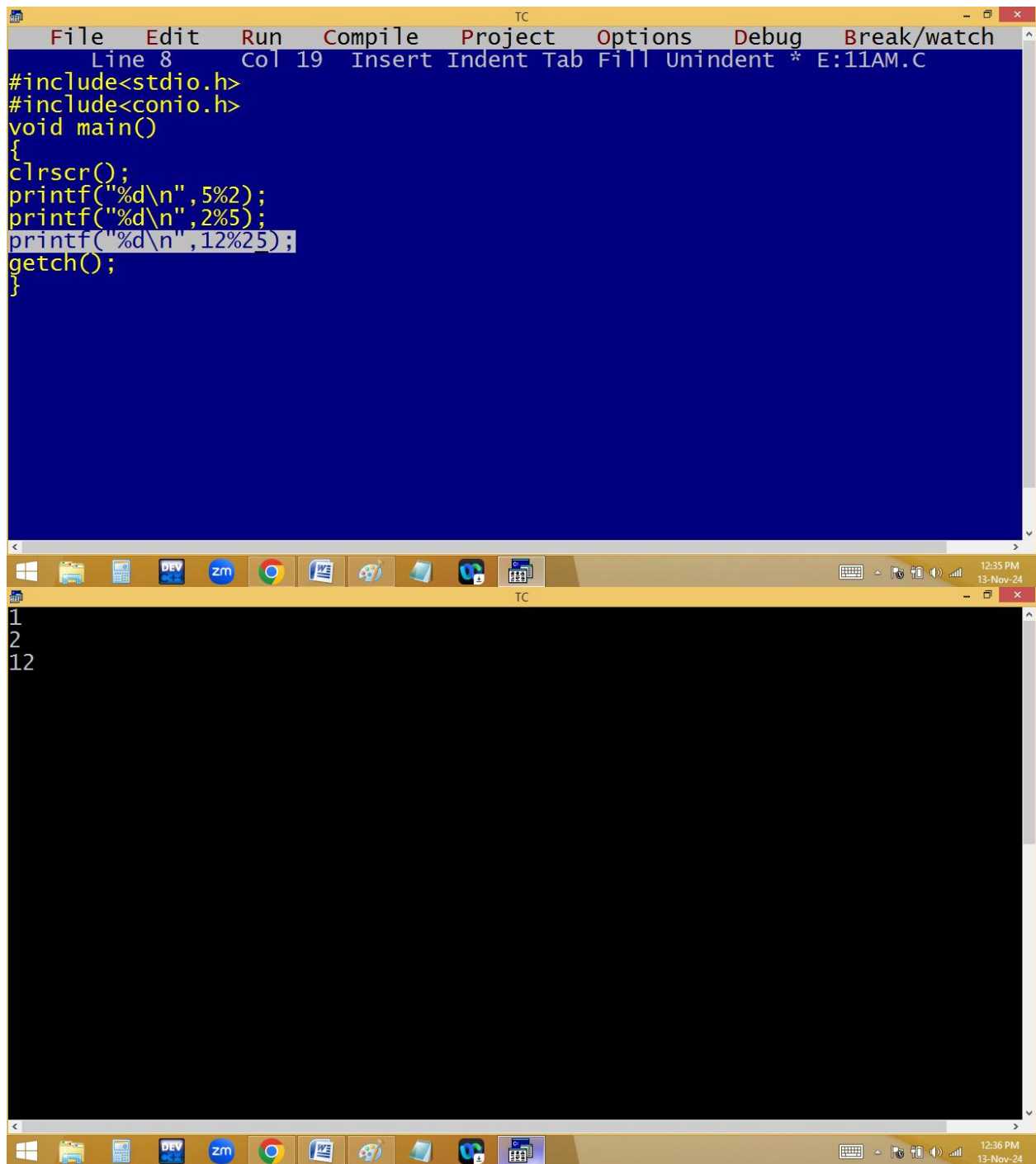
Eg:

$$5 \% 2 = 1$$

$$\begin{array}{r}
 2 \overline{) 5} \quad 2 \text{ } \\
 \underline{4} \\
 1
 \end{array}
 \begin{array}{l}
 \text{Quotient} \\
 \text{Remainder} \%
 \end{array}$$

$$2 \% 5 = 2$$

Note: If the divisor is bigger than dividend then the dividend is the answer.



The image shows a screenshot of the Turbo C++ (TC) IDE. The top window displays a C program with the following code:

```
File Edit Run Compile Project Options Debug Break/watch
Line 8 Col 19 Insert Indent Tab Fill Unindent * E:11AM.C
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d\n",5%2);
printf("%d\n",2%5);
printf("%d\n",12%25);
getch();
}
```

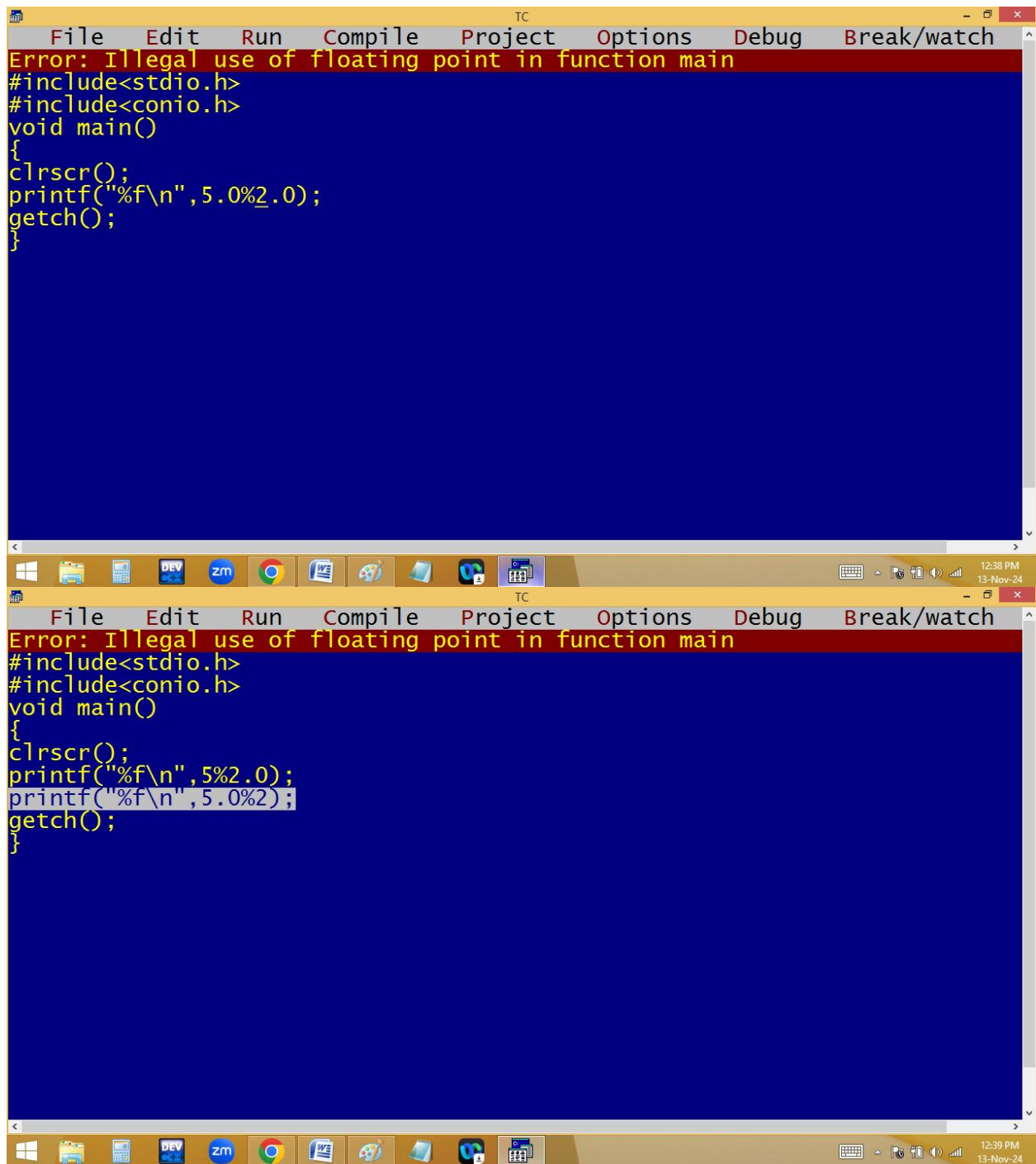
The bottom window shows the output of the program, which is:

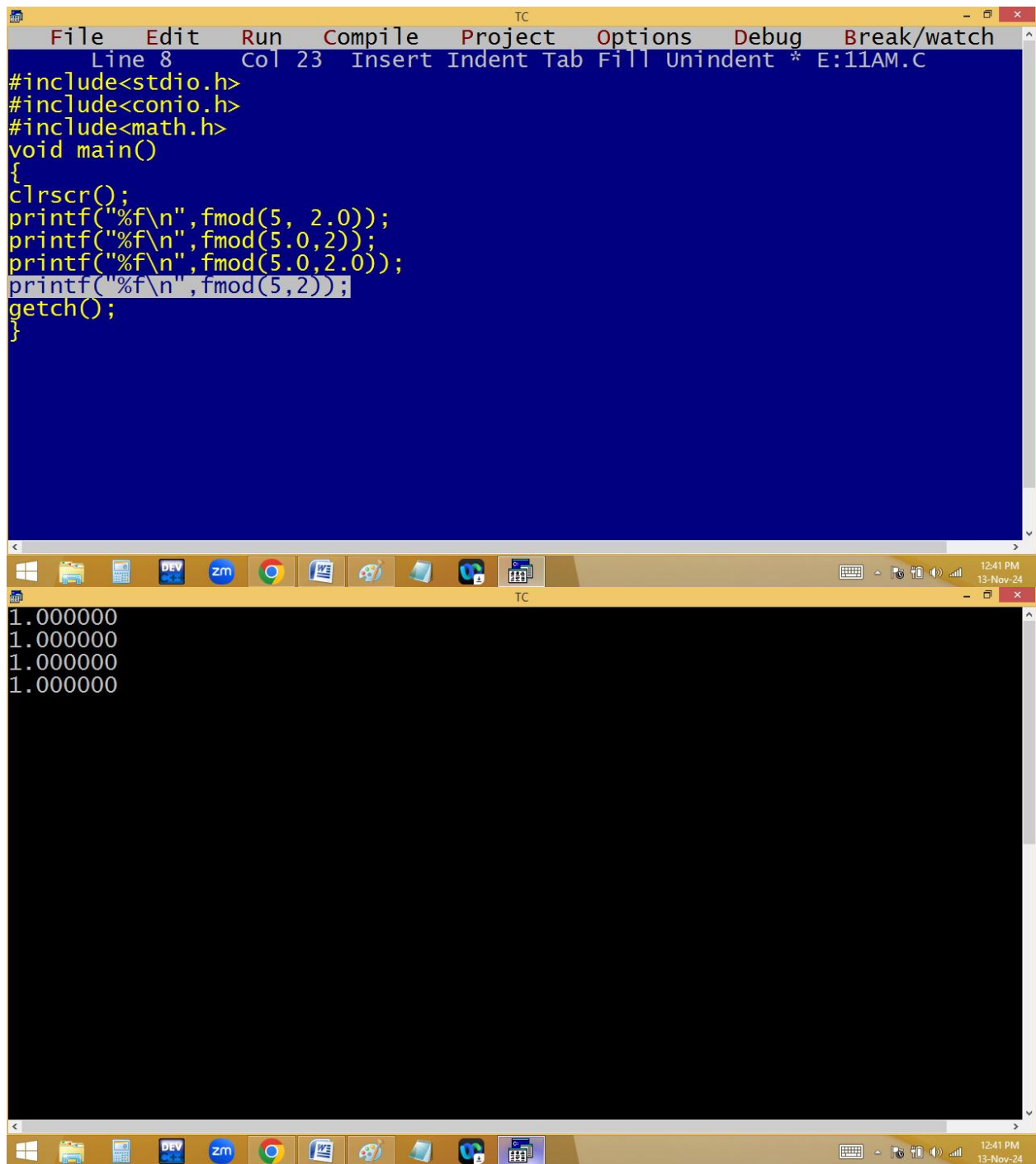
```
1
2
12
```

The Windows taskbar at the bottom shows the time as 12:35 PM on 13-Nov-24.

5.0 % 2.0 = Error

Note: We can't do floating modules with **%** operator. For this we have to use **fmod()** available in **<math.h>**





The image shows a screenshot of the Turbo C++ (TC) IDE. The top window displays a C program with the following code:

```
File Edit Run Compile Project Options Debug Break/watch
Line 8 Col 23 Insert Indent Tab Fill Unindent * E:11AM.C
#include<stdio.h>
#include<conio.h>
#include<math.h>
void main()
{
clrscr();
printf("%f\n",fmod(5, 2.0));
printf("%f\n",fmod(5.0,2));
printf("%f\n",fmod(5.0,2.0));
printf("%f\n",fmod(5,2));
getch();
}
```

The bottom window shows the output of the program, which consists of four lines of floating-point numbers:

```
1.000000
1.000000
1.000000
1.000000
```

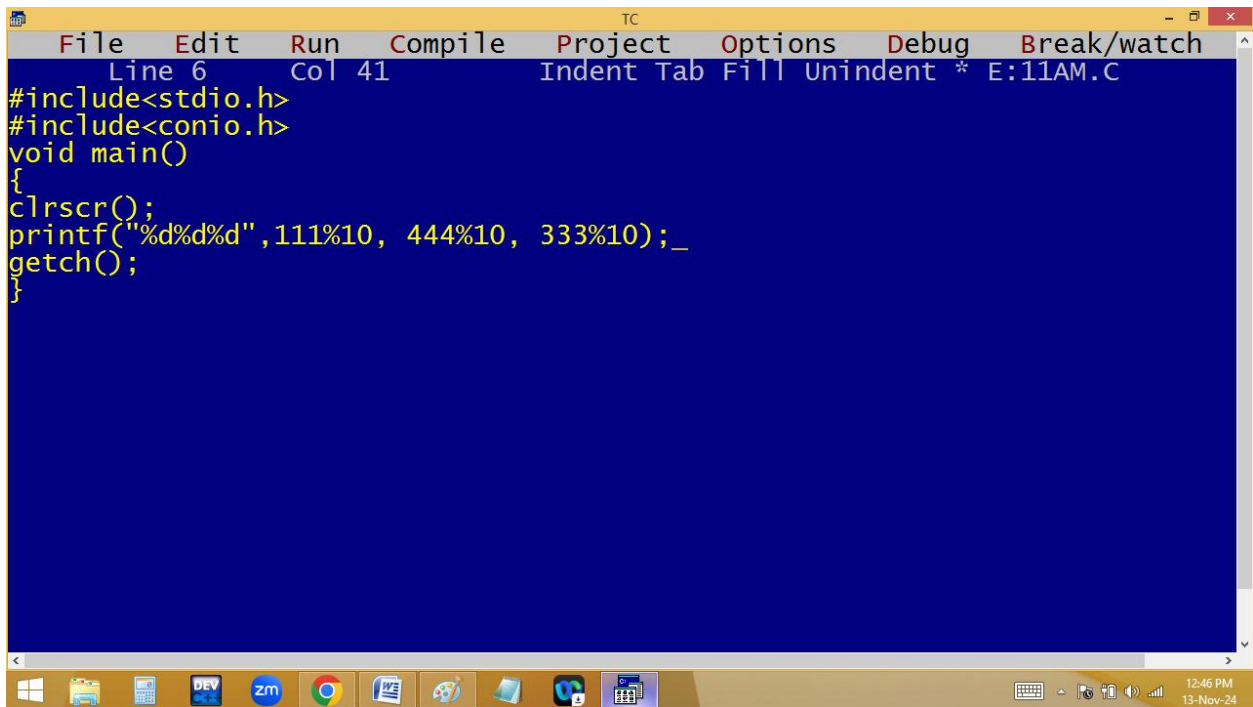
The Windows taskbar at the bottom indicates the time is 12:41 PM on 13-Nov-24.

111%10=1

444%10=4

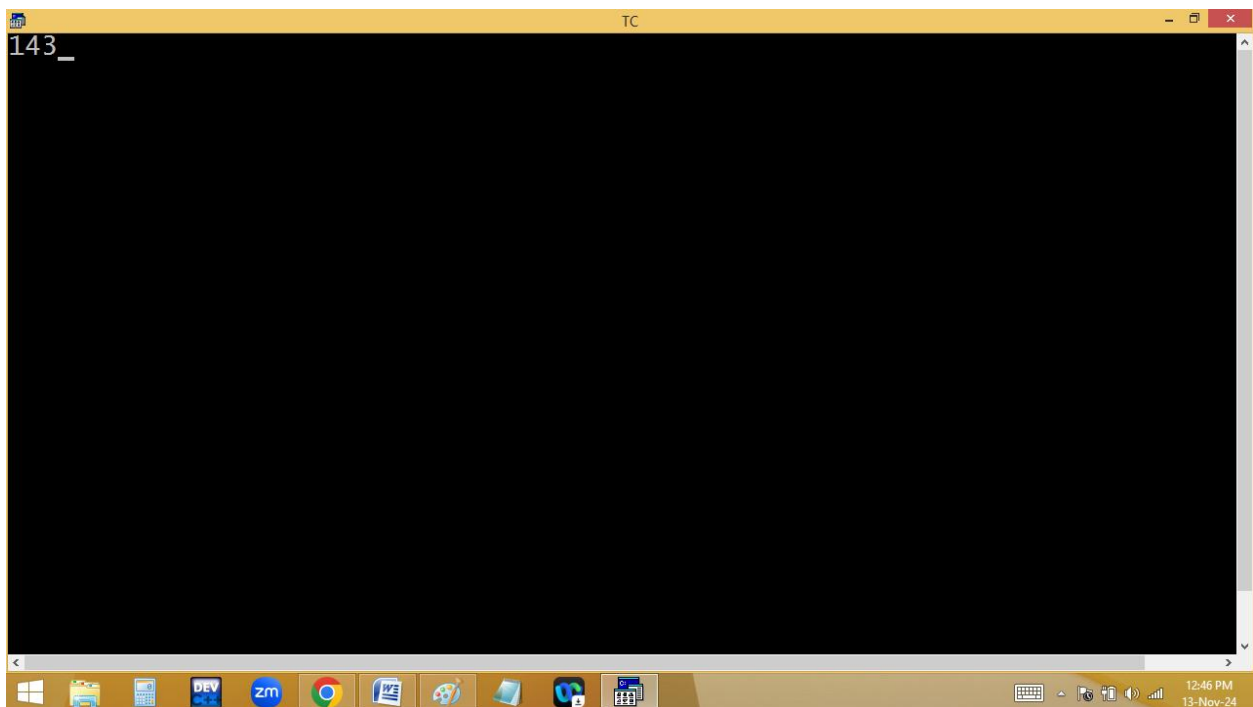
333%10=3

Note: Any $\text{no} \% 10$ gives last digit.



The screenshot shows the Turbo C++ IDE with a blue background. The menu bar includes File, Edit, Run, Compile, Project, Options, Debug, and Break/watch. The status bar at the bottom indicates Line 6, Col 41, and the file name E:11AM.C. The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d%d%d",111%10, 444%10, 333%10);_
getch();
}
```



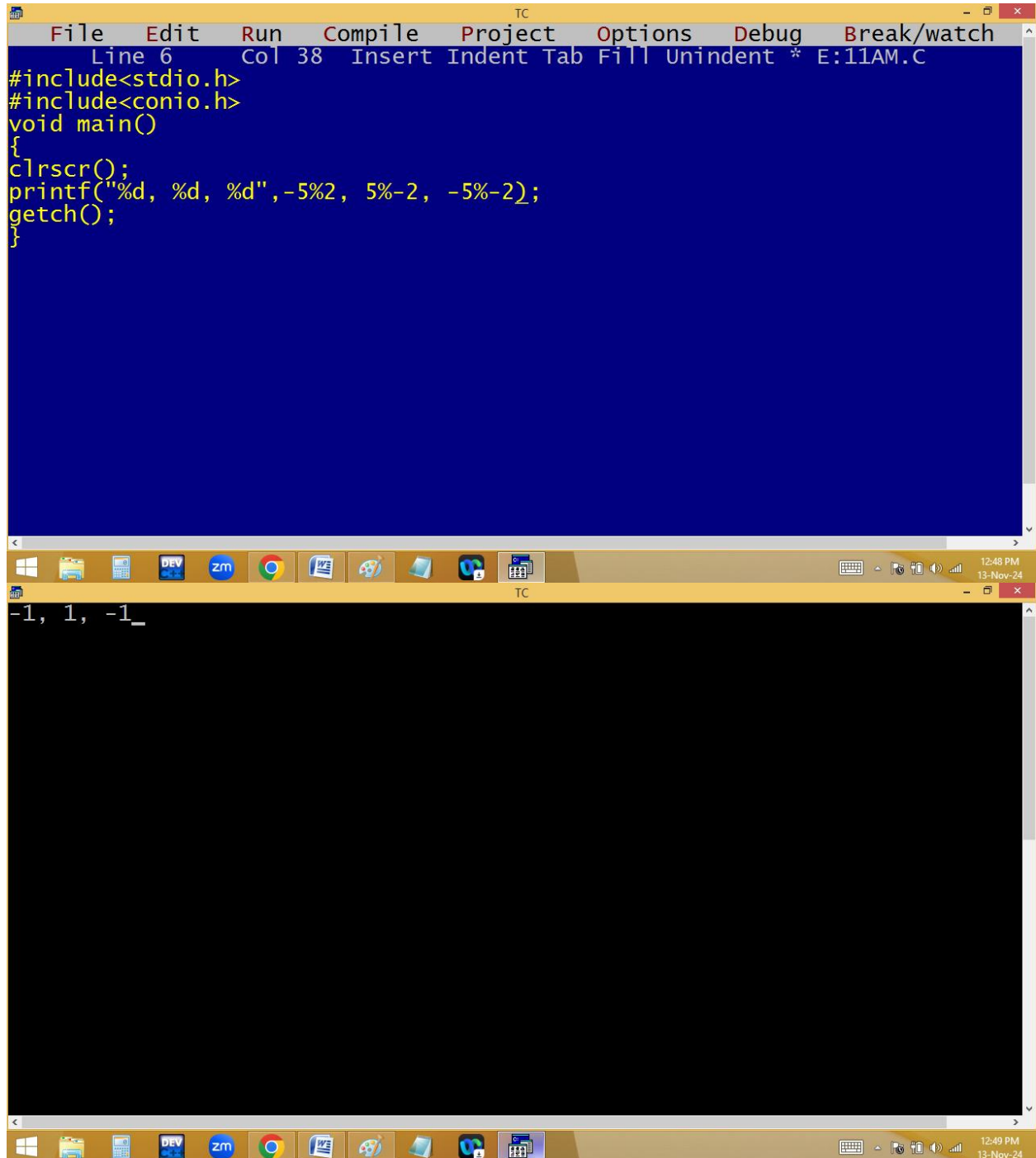
The screenshot shows the Turbo C++ IDE with a black background. The output of the program is displayed in the top left corner as "143_". The status bar at the bottom indicates the time 12:46 PM and the date 13-Nov-24.

$$\text{-5 \% 2 = -1}$$

$$\text{5 \% -2 = 1}$$

-5%-2= -1

Note: if the numerator is negative then result also negative.



The screenshot displays the Turbo C++ (TC) IDE interface. The top window shows a C program with the following code:

```
File Edit Run Compile Project Options Debug Break/watch
Line 6 Col 38 Insert Indent Tab Fill Unindent * E:11AM.C
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d, %d, %d",-5%2, 5%-2, -5%-2);
getch();
}
```

The bottom window shows the output of the program:

```
-1, 1, -1_
```

The Windows taskbar at the bottom indicates the time as 12:48 PM and 12:49 PM on 13-Nov-24.

